

Detection and Prevention of Lithium Plating in Lithium-Ion-Batteries

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”Die Brillianz des Zitierten färbt selten
auf den Zitierenden ab.”
(Heribert Herrlich)

Danksagung/Vorwort

Abstract

Abstract in Englisch (if applicable)

Kurzzusammenfassung (Abstract)

Kurzzusammenfassung auf Deutsch

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Introduction and Motivation

In the last decade, lithium-ion batteries (LIBs) have consolidated their prime position as the go to battery for all consumer electronics [Zubi.2018]. With the climate protection targets and the demand for a rising amount of electric vehicles for private and public transport, the LIB is also the only technology that has the needed energy density, as well as the development level necessary, to support the transition from fossil fuels to CO₂ free transportation. While there are new technologies developed with even higher theoretical energy densities, none of the candidates has reached a technology readiness level yet that could challenge the usability of the LIB [Placke.2017]. One big concern with batteries in general however is the ongoing degradation through battery ageing during operation that not only increases the running of costs of all electric vehicles (EVs) but also reduces the customer acceptance of the vehicles, which is vital for a widespread change to electric transportation [Agubra.2013]. With increasing ageing, not only the capacity of a cell fades but also the available power. As of time the main efforts in the battery research are targeted towards increasing the energy density to enable higher range on full electric vehicles and to reduce the ageing on the cell to increase the overall lifetime [Vetter.2005, Lin.2015]. One of the strongest ageing effects possible in LIBs is the so call lithium plating, which describes the unwanted deposition of metallic lithium on the anode surface during charging. This deposition takes place when the device is charged at very low temperatures, or when very high charging currents are applied. The capacity losses of plating can lead to the end of life of a cell within 10s of cycles, when the nominal cell life can last for hundreds or even thousands of cycles [JiangFan.2006]. The plating process can also cause the growth of lithium dendrites, small lithium metal structures that can pierce the separator and cause shortcuts within the cell. These can lead to cell failure or in the worst case to a thermal runaway of the cell, a process where the cell reaction is self-energizing and causing a cell fire [Waldmann.2018].

The conditions for lithium plating are rarely met for cells within consumer electronics, since they are normally charged at moderate rates within living areas. This is different for EVs however. Electric cars are supposed to work under all normal outside conditions, charging at low temperatures is something that cannot be prevented. Today the vast majority of customers with electric cars also own a house or have access to a garage to charge their cars at higher temperatures under better [Nobis.2019]. With the target of a much higher percentage of EVs in the future however, this cannot be guaranteed for each owner of an electric car and some will

have to be charged at public charging stations outside. To compensate for this effect, it is possible to equip EVs with a heating system that heats up the battery to the optimal charging temperature prior to the charging process. The drawback of this method is for one the energy losses induced by the necessary heating process that increase the running costs of the car and the increased charging time due to the added heating time. In some cases there may not even be enough energy left in the cell to heat up properly to prevent plating during charging. The batteries are heating up due to the ohmic losses during the charging process, but this does not fully prevent the ageing caused by plating. Another aspect that is of vital importance for EVs is the charging speed [Farjah.2018]. To compete with combustion engine cars the charging speeds of EVs would have to be in the range of minutes. Achieving this would greatly increase the acceptance of the EVs as a fully capable replacement. This would however require charging speed that greatly exceed the recommended nominal charging speeds for standard commercial LIBs. To fully charge a battery within ten minutes a charging speed of 6 times higher than average expected data sheet speeds is required. It is possible to manufacture cells that can handle higher charging speeds, but this is always at the cost of energy density, which is vital for the range of the EV. For these very high charging speeds, plating effects are to be expected even at moderate temperatures, leading to very fast cell degradation [Waldmann.2018].

Grid stabilization using EVs could be the solution for the strongly fluctuating energy output of renewable energy sources. All EVs that are not currently in use and that are connected to the power grid could be used to stabilize the grid and store excess energy in times of overproduction, while at the same time providing energy in times of high demand [Nunes.2017, Pramana.2017, Quinn.2010, Yan.2014]. If the vehicles are used in this way, low environment temperatures could be very damaging for the battery cells. In this case, the energy influx is unpredictable and uncontrollable and thus no prior heating is possible. Always keeping the batteries at a safe temperature however would produce very high energy losses.

The amount of available spent batteries is going to increase significantly within the next few years when the first generations of car batteries will reach the end of life. The plating effect is here also attributed to sudden cell death at very low state of health [Waldmann.2018], when the surface electrolyte interface (SEI) layer is getting very thick. Predicting this mode of cell failure could be very beneficial for second life usage of spent EV batteries.

Understanding and preventing plating could not only increase the general lifetime for LIB, but also enable the usage of EVs as a grid support for renewable energies, greatly increasing the usability of fully renewable energy sources to reduce CO₂ emissions. It also can help make second life applications more reliable and thus extend the commercial value of the batteries. At the same time, it increases the safety of LIBs by preventing dendrites that can cause cell failure or dangerous thermal runaways. A deeper understanding of the plating process can also help to increase the usability of metallic lithium in future battery technologies. The usage of metallic lithium as the anode in a lithium battery drastically increases the

energy density. The re-deposition of the lithium on the anode during the charging process in the battery however is hard to control and leads to very low cycles life of those cells [**Lin.2017**, **Cheng.2017**]. Even for completely new technologies, the usage of lithium is highly likely [**Placke.2017**]. lithium is not only the lightest solid material in existence but it also has the highest negative electro potential. This enables very lightweight batteries with a high voltage and thus high energy density.¹ {x}

¹Footnote example

2.1 Lithium-Ion Batteries

LIBs are using lithium-ions to store energy. Contrary to the lithium-metal battery (LMB), where lithium exists in its elemental state inside on the battery, the lithium in the LIB only exists in ion form, intercalated or in a chemical compound. Although the LIB has a lower possible energy density compared to the LMB, the LIB became the standard of lithium batteries. The first lithium batteries on the market in the late 1980s used lithium metal as the anode, but following a series of accidents caused by the re-deposition of the lithium during charging and the resulting risk of lithium dendrites growing on the surface, these cells had to be recalled [K.Brandt.1994]. The subsequent failure to solve the safety problems of the LMB quickly led to the discovery of alternate anode materials that were able to store the lithium-ions in a safer and more reliable way. The LIB is not only safer than the LMB, it also possesses a much higher cyclability due to the more controlled lithium deposition process on the anode. The LIB quickly took over the market for consumer electronics, completely replacing the leading technology at that time, the nickel-based batteries, due to its superior energy density and cycle life. The introduction of this technology enabled a lot of the mobile devices that are a natural part of our lives today: smartphones, tablets, cameras, smartwatches and so on. As of now the LIB is also the only commercially ready technology that can enable full electric transport for private use and is thus an integral part of the worldwide efforts to become free of fossil fuels and reduce the overall CO₂ emissions. And even though the demand for high energy density in the sector of stationary energy storage is naturally not as high as for mobile applications, LIBs are starting to see usage in large scale stationary storage for grid stabilization as well [Guthrie.2020].

2.1.1 Components and Functionality

The LIB is, like all electrochemical storage systems, made up primarily of an anode and a cathode, that drive the redox-reaction. Figure ?? shows the general schematic of a LIB and all its components. In most commercially available LIBs today, the **anode** is mainly made of Graphite (C₆). This graphite and added binders are pressed onto a copper foil that functions as the current collector, while the **cathode** is a lithium-metal-oxide (LiMeOx) pressed on aluminum foil.

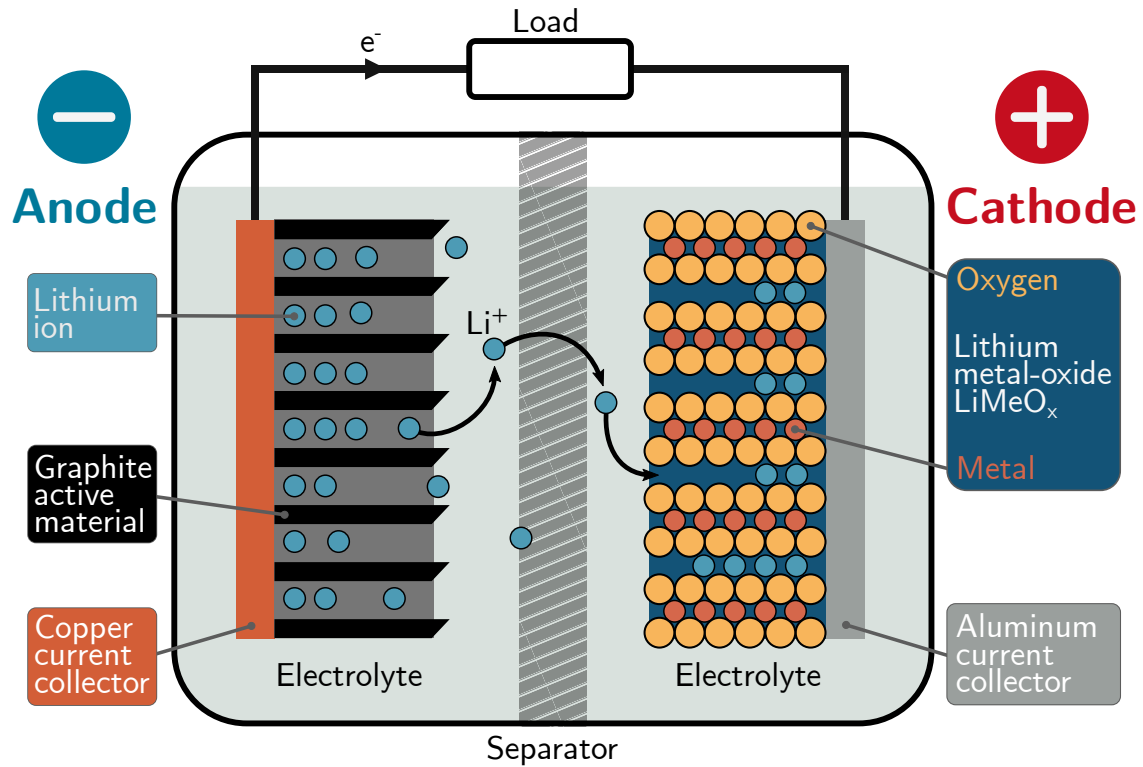


Figure 2.1: Schematic of a general LIB

In the LIB the lithium-ions are responsible for the storage of energy. During charging and discharging the ions are moved from the anode to the cathode and vice versa. It is not necessarily always the case in all electrochemical systems that the main ion is moved during the reaction. Even though there always has to be an ion moving inside of the cell to close the electrical circuit, e.g. in lead-acid batteries the moving ion is a hydrogen ion, not a lead ion.

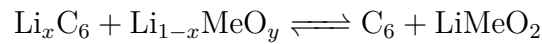
The **electrolyte** in LIBs is not taking part in the cell reaction as it was the case in its technological predecessors, the nickel-hydroxide and lead-acid batteries, but is solely in the cell for the ion transport. For this reason the amount of electrolyte in the cell can be reduced to a minimum necessary to properly connect all the cell elements. Normally the electrolyte of commercial LIBs is composed of an organic solvent with a lithium salt as the lithium-ion source. The usage of a aqueous electrolyte in cell cells containing lithium is not possible, since lithium reacts with water. Typically the electrolyte is composed of a mixture of organic solvents to get the best performance. The most common solvents used are a combination of ethylene carbonate (EC) and dimethyl carbonate (DMC) with lithium hexafluorophosphate (LiPF_6) as the lithium containing salt. This combination has proven to have a good combination of temperature stability, good ion conductivity, good wetting ability and not co-intercalation. In the first LIBs models propylene carbonate (PC) was used, a remainder of the short-lived LMB period, having a even better lithium-ion conductivity. This however was causing quick anode degrada-

tion when the solvent molecules undesirably intercalated into the graphite together with the lithium ions [K.Brandt.1994]. The electrolyte also plays a mayor role in the formation of the surface electrolyte interface (SEI) layer, that impacts the cell performance as well as the cell degradation and will be explained in detail in chapter ??.

The **separator** is an element in any electrochemical cell that is physically separating the the two electrodes whilst electrically isolating them against each other. At the same time the separator must be permeable for the ion current inside of the cell, which in most cases means a porous separator that is permeable for the electrolyte. In LIBs the most common separator is a polymer membrane, e.g. made of polypropylene (PP) or polyethylene (PE). The desired properties of the separator are a low weight, high thermal and mechanical stability and low material costs [Deng.2015].

Charging and Discharging

In a fully charged LIB all lithium-ions are intercalated inside of the anode. Intercalation means that the ions are positioned in between the graphite layers, but no chemical reaction has taken place. It is described by a reversible process that is keeping the structural integrity of the intercalation host material [Li.2019]. The fact that there is no chemical reactions is one factor of the high lifetime of LIBs. Since there is no new compound forming, the stress due to rearrangement of the material during the charging and discharging process is very low. The graphite is increasing in size by less than 10% during the lithium-ion intercalation [Winter.1998]. When the battery is discharged the lithium-ions de-intercalate from the anode by releasing one electron becoming Li^+ and move through the electrolyte to the cathode, where they are reacting with the metall-oxide to form a lithium-metal oxide. The cell total cell reaction in the discharge direction can then be described as:



Where the x describes the ratio of lithium remaining within the anode, this number is equivalent to the state of charge (SOC). The graphite can have different lithiation-states that can be seen in plateaus in the open circuit voltage (OCV) [Bauer.2016]. The Me is representative for a group of metals that can be used in the cathode.

During the charging process, this reaction is simply inversed and the lithium-ions are moving from the cathode to the anode, where they are intercalating into the graphite and take in one electron.

Since LIBs are very sensitive to over- and undercharging, it is very important to stay within the voltage limits for the given material combination. For this reason the most common choice of charging method for LIBs is the constan current constant voltage (CCCV) charging method. Using this method, the cell is charged with a

constant current until the end-of-charge voltage is reached (typically 4.2 V) and then that voltage is held during a constant voltage phase while the current drops. When the current reaches a minimum current threshold, the charge is finished. The minimum current is necessary due to self discharge and parasitic currents, because of which the charging current can never reach zero in a real application. Figure ?? shows the voltage, current and SOC curves for CCCV charging, where I_{ch} is the charging current and I_{thres} the threshold current at which the charging process is terminated. V_{max} and V_{min} denote the end-of-charge and end-of-discharge voltage window of the battery, respectively.

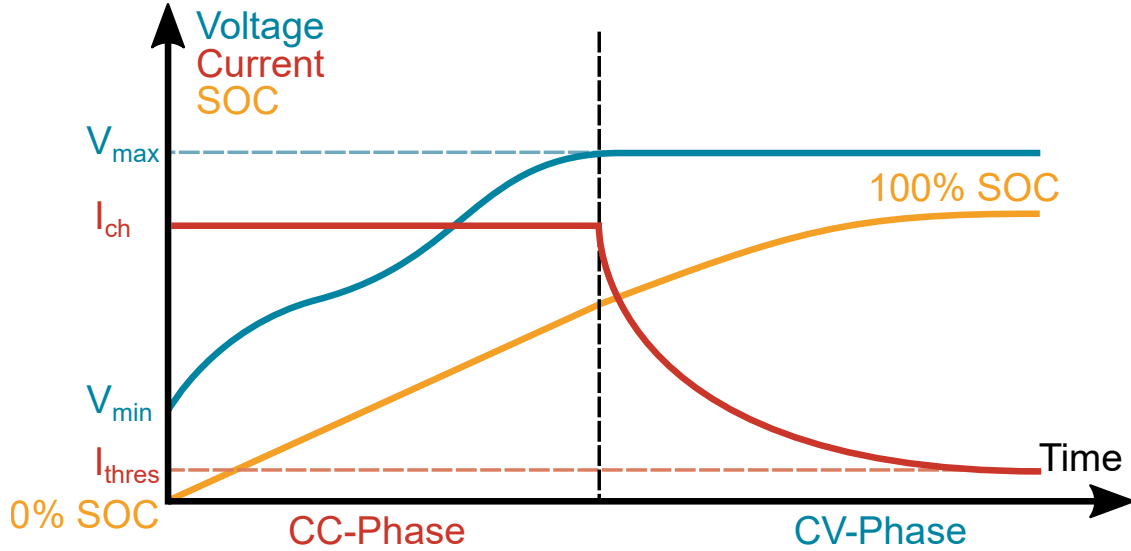


Figure 2.2: Voltage, Current and SOC curves of a typical CCCV charging process

During the CC-phase the majority of charge is put into the battery and the SOC is raising linearly. In most cases the battery is charged to around 80% SOC in the CC-phase. The CV-phase then charges the remaining amount to 100% SOC. The CC-phase makes up around half of the total charging time. For this reason a lot of fast-charging methods only use the CC-phase for charging and don't use the full battery capacity.

2.1.2 Anode

Graphite is currently the most used anode material for lithium-ion batteries due to its high practical capacity and low potential vs Li, enabling it to store a high amount of energy, and a rather flat discharge curving, giving a constant power output over a big range of the capacity [Li.2019]. The anode of a commercial LIB as of today is in most cases made of a slurry of graphite (C_6) with added binder and carbon black that is pressed on a copper foil. The binder used is typically polyvinylidenefluorid (PVDF) [Wang.2017]. The carbon black is added to increase the electric conductivity of the active material. The carbon has the shape of particles with varying sizes depending on the producer. Differently sized

particles can have different effects on the cycle-lifetime and other parameters of the battery performance, like the available power density.

The anode is typically oversized compared to the cathode to prevent lithium plating at the edges [Agubra.2013]. If the electrodes had the same size the increased electrical field strength at the edges of the anode could lead to lithium plating in these regions. This is sometimes also called the anode overhang.

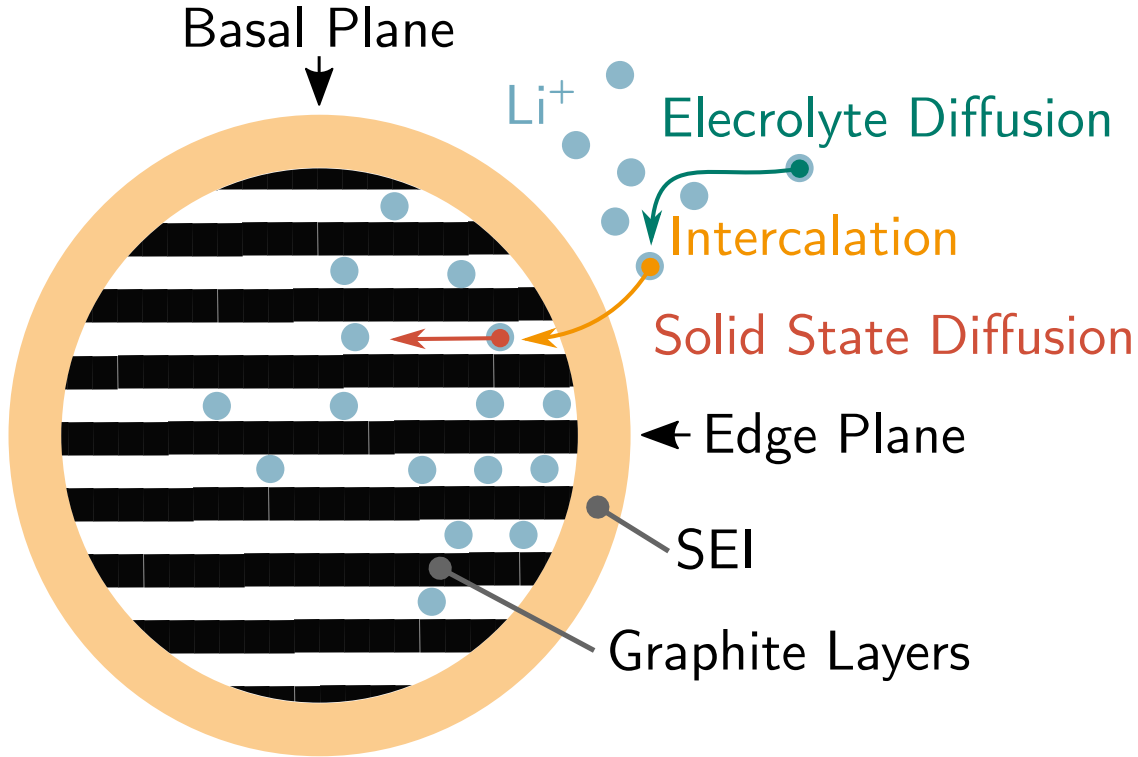


Figure 2.3: Schematic view of one graphite particle inside the anode active material

All graphite particles also consist of basal and edge planes. The edge planes are perpendicular to layer direction of the graphite crystals. They provide an easy entry point for the lithium-ions to intercalate into the Graphite. This is why most of the lithium-ion insertion and diffusion into the active material starts from the edge planes. The basal planes are along the layer direction of the graphite crystal and lithium-ions cannot intercalate across a basal plane into the graphite, which is why they are the less reactive areas of the graphite particles. The ratio of edge planes to basal planes has a high impact on the electrochemical performance of a given graphite material [An.2016]. The same way the orientation of the particles has a similar influence, with less oriented graphite generally having a worse performance due to a more difficult diffusion for the lithium ions [Agubra.2013].

In the charged state the lithium-ions are intercalated into the graphite, located between the graphite planes. Lithium can naturally diffuse into graphite and is evenly distributing in the anode active material over time after the charging in

finished. The resulting material of graphite with intercalated lithium-ions is called a graphite intercalation compound (GIC) that is characterized by a mixed ionic and electric conduction [Li.2019].

The volume expansion of graphite (C_6) compared to fully lithiated graphite (LiC_6) is approximately at about 10% to 13% when the lithium-ions inhabit the free lattice spots within the graphite [Lin.2016, Li.2019]. This effect is quite low compared to other materials, but still leads to mechanical stress inside of the cell during cycling.

To avoid lithium plating due to anode overcharging, explained in more detail later in chapter ??, the anode is usually oversized by around 10% compared to the cathode [Gallego.2014].

Anode Ageing

In general ageing of all cell components can be divided into calendar ageing and cyclic ageing. Calendar ageing describes the chemical degradation over time due to unwanted mostly unavoidable side reactions. Cyclic ageing on the other hand describes the stress on the cell components during cycling, i.e. in case of a LIB when the lithium-ions are shifted from anode to cathode and vice versa. For calendar ageing the main influencing factors are the temperature and the SOC of the cell. The result of battery ageing is typically loss of usable capacity and an increase of the internal resistance, leading to a lower available power density [Lin.2015]. Depending on the occurring ageing mechanisms, there can be primarily capacity loss, primarily an increase in internal resistance or both.

For the anode the main ageing mechanism is the formation, growth and reformation of the SEI layer, that forms on the surface of the graphite particles. This layer consumes lithium ions, reducing the available lithium inventory and at the same time increase the cell impedance. A more detailed description of the ageing mechanism can be found in chapter ??, since this layer also has a big impact on the lithium plating effect [Lin.2015].

Another ageing effect is the mechanical stress the anode receives during charging and discharging due to the change in volume of the cell. While the volume change of around 10 to 13% is rather small, prolonged cycling can lead to contact loss between the graphite particles, the active material and the current collector and to a change in porosity of the anode material [Vetter.2005]. Other sources of mechanical stress can be the co-intercalation of solvent ions, that can cause graphite exfoliation and gas evolution during electrolyte decomposition [Lin.2015]. These mechanical ageing effects can affect the capacity, due to loss of active material or contact loss with the active material, or internal resistance due to longer diffusion paths [Vetter.2005]. Over time the pores of the anode material can also be clogged with solvent decomposition products, leading to a higher charge transfer resistance [Agubra.2013].

The strongest effect on the capacity loss has the lithium plating, explained in more detail in chapter ??. Here an unwanted deposition of metallic lithium on

the anode surface can happen under specific conditions, including charging at low temperatures and high charge currents [Vetter.2005].

Overall the capacity fade even under the same conditions are not linear with time can change depending depending on those conditions [Barre.2013].

The SEI Layer

The SEI layer does not only have a huge impact on the cell performance, it also is the main reason for cell ageing in LIBs [Vetter.2005]. The SEI is a thin layer on the anode surface that is forming due to electrolyte decomposition. There is no electrochemically stable electrolyte for the operating voltage range of a LIB, so a decomposition is unavoidable. Due to the high reactivity of water with lithium an aqueous electrolyte is no option in the current state LIBs. The working voltage window in the battery will either cause electrolyte decomposition at the anode or cathode at the respective voltage values of a fully charged or fully discharged cell [Wang.2018, Lin.2016]. The stability window for most of the commonly used electrolyte parts (e.g. EC, DMC) is from around 0,8 V to 4,7 V vs Li/Li⁺ [Goodenough.2010, Palacin.2018]. This window is further decreased by the cell temperature leading to electrolyte degradation also at the cathode for elevated temperatures [Shikano.2007, Rahman.2007]. As most commercially available LIBs operate from around 0,2 V vs Li/Li⁺ when fully charged to 1,7 V vs Li/Li⁺ when fully discharged, the decomposition takes place on the anode side at high SOC. In the decomposition process not only the components of the electrolyte are consumed, but also the lithium-ions present in the electrolyte. This does reduce the total number of available lithium in the cell and in consequence the capacity of the cell. This implementation of lithium-ion however into the SEI does enable lithium-ions from the electrolyte to pass through it to reach the anode surface. Without the lithium-ions in the SEI the cell could not function since the lithium-ions from the electrolyte could not intercalate into the anode surface. The decomposition reaction can also take place at voltages inside of the stable window, but in this case it was shown that the reaction is reversible since the electrolyte molecules stay intact [Seidl.2016].

Since the SEI is electrically insulating, the growth of the layer should in theory stop after its initial formation. In a real cell however there is always a small current due to electron hopping leading to a slow but constant SEI growth over time. This growth is accelerated with higher temperatures. During cell cycling the SEI can also be damaged due to the swelling and contraction of the anode during the lithium intercalation and de-intercalation, facilitating a reformation of the SEI at newly exposed anode surface. There can also be co-intercalation of other electrolyte components and gas evolution during the decomposition process causing extra stress on the SEI, with the highly flammable gas additionally posing a safety risk [Seidl.2016, Balagopal.2018]. Due to the fact that the decomposition of existing SEI is different from the reaction on the anode surface, the mechanisms changes with a higher coverage of the electrode surface [Wang.2018]. The SEI is

not only gradually reducing the capacity of the cell but also having a major impact on the self-discharge characteristics, achievable power density and safety of the cell [An.2016].

These effects lead to a number of desired parameters that can create a stable SEI that can lead to a long cell life. The SEI should be as thin as possible while at the same time being highly electrically insulating to prevent further decomposition. Even if the material composition of the layer is technically electrically insulating, it needs to be thick enough to block electron tunneling [Lin.2016]. But it should also be permeable selectively for lithium-ions for a low cell internal resistance. The layer should also be chemically inert towards all other cell components at a wide range of potentials and temperatures to prevent dissolving of parts of the SEI back into the electrolyte. To compensate for the change in anode volume during cycling it should also have a good mechanical stability to prevent breaking and subsequent reformation [An.2016, Wang.2018]. The fracturing of the SEI is even reported to have a bigger effect on the capacity fade than the cracking of the underlying graphite particles due to the fact that it has a higher tendency to fracture [Deshpandezand.2017].

To achieve these desired SEI attributes, the cells are normally charged under special conditions by the manufacturer during their first charging cycles. This process is called cell formation. Before the formation process a full coverage of the anode with the electrolyte has to be assured. This is done by the wetting process, which is typically a prolonged period of resting time (sometimes at elevated temperatures) following the electrolyte filling. During the first cycle of the formation 10%-20% of the whole cell capacity are typically consumed for the SEI generation [Patil.2008, Lin.2016, Gallego.2014]. The factors that the manufacturer can directly control during the formation process are the applied current, the different cut-off voltage levels and the temperature. These all can have a strong influence on the parameters and stability of the formed SEI, creating different microstructural and chemical properties that can lead to a different long-term chemical and mechanical performance [An.2016]. The current during the formation is typically kept at very low C-rates, ranging from C/5 to C/20 [Wood.2015]. Most of the SEI is theorized to be formed at 0.2 V to 1 V vs Li/Li⁺ [An.2016]. Other parameters that can be adjusted in cell design that also influence the SEI formation are the graphite type, graphite morphology, electrolyte composition and electrolyte additives, but also factors like particle size of the graphite, basal-to-edge-plane ratio, pore size, degree of crystallinity, Li⁺ diffusion rate and chemical composition of the surface (e.g. impurities) can affect the SEI composition [An.2016, Seidl.2016, Bhattacharya.2012]. A commonly used additive is vinylene carbonate (VC), preventing exfoliation during lithium intercalation in the beginning of the SEI formation [Aurbach.1998, Flandrois.1999, Tasaki.2006]. Using bigger particles can lead to a smaller capacity loss during SEI formation since the surface to volume ratio is lower in this case, but these particles are also more prone to exfoliation [An.2016]. The whole wetting and formation process can take up to 0.5 to 2 weeks [Wood.2015] and needs dedicated cycling capacities for each single cell, make it

one the most important bottlenecks of cell production.

The resulting layer has a thickness in the range of nano meters, with reported values going from 3 to 120 nm [An.2016, Bhattacharya.2012, Shinagawa.2017, Seidl.2016], but other publications also report widely varying thicknesses in the micrometer range [Bhattacharya.2012, Balagopal.2018]. Since the intercalation reaction happens on the edge planes of the graphite particles, the potential on these planes is higher than on the basal planes. This leads to a thicker and differently composed SEI on the edge planes [Seidl.2016]. The thickness of the SEI is also varying with the position of the electrode, with the SEI generally being thinner towards the center, and dependent on the type of aging, being thicker when it is caused by calendar ageing [Mussa.2019]. All these factors lead to a differing lithium diffusion constant throughout the SEI in different locations [Bhattacharya.2012]. The pressure on the layer also influences the electron tunneling ability, with more tunneling when the layer is under tension and less tunneling if the layer is under pressure [Lin.2016].

The chemical composition of the SEI layer is also highly dependent on the conditions under which the layer is formed. A loose organic SEI with rather low conductivity for the lithium-ions is formed at higher charging anode potentials and more compact inorganic structures with higher ion conductivity at lower potentials [Bhattacharya.2012, Tran.1996]. The charging rate on the first formation cycles on the other hand affects the porosity of the layer, with a more porous layer and higher resistance at higher charging rates. Water residue in the graphite can cause HF creation in electrolytes with LiPF_6 , which is the most commonly used salt in modern lithium ion batteries. The HF can then attack the SEI and cause a drop in performance of the LIB [An.2016]. The overall structure of the SEI can roughly be described as a multi-structure bilayer [An.2016, Wang.2018, Bhattacharya.2012, Lin.2016]. The edge sites mainly contain a loose layer filled with electrolyte of inorganic lithium carbonates, organic lithium alkali carbonates and polymeric compounds on the side facing the electrolyte, while the graphite site is a more dense layer of LiF, Li_2O and Li_2CO_3 [Peled.1998, Eshkenazia.2004]. Between these two phase a mixed layer can be found. In general it can be said, that the SEI compounds closer to electrode are reduced stronger than the compounds in contact with the electrolyte [Seidl.2016]. On average more than half of the SEI is composed of LiF and Li_2CO_3 . Compared to the edge sites, on the basal site a higher concentration of lithium carbonates can be found on the electrolyte site, and a lower content of LiF [Peled.1998, Eshkenazia.2004]. The whole composition is on the basal sites is also less organic [Seidl.2016]. To prevent the loss of the SEI due to re-dissolution into the electrolyte [Wang.2018], it is desired to create a layer mainly composed of inorganic components like Li_2CO_3 rather than ROLi or ROCO_2Li . The ROCO_2Li can react with traces of CO_2 or H_2O present to form lithium carbonate [Lu.2008]. Hao et al. [Hao.2017] also notes, that the lithium ion concentration in the SEI highly fluctuates due to an heterogeneous SEI structure. The micro structure is also reported to contain nano-crystalline domains of 5-20 nm in size manly consisting of Li_2CO_3 and Li_2CO_2 [Bhattacharya.2012]. In

some cases elements from the cathode can cross over to the anode and cause SEI growth, as shown for Mn_2^+ ions [Leung.2017].

The analysis of the SEI is very hard due to its very small layer thickness and sensitivity to oxygen and moisture [An.2016]. To get a deeper understanding of the SEI a number of different measurement and modeling approaches have been used, including ab initio molecular dynamics (AIMD) simulations [Soto.2018], to get insight into the reaction dynamics, [Shinagawa.2017] et al. using a monte carlo simulation to determine the relationship between the layer thickness and its lithium ion conductivity, a mathematical model proposed to describe the mechanical stress on the SEI and subsequent fracturing during cycling [Deshpandezand.2017], while atomic force microscopy (AFM) was used in [Bhattacharya.2012] to measure the thickness and coverage of the SEI.

The continuous growth of the SEI layer over the lifetime of the cell depends on a lot of factors. Firstly a higher cell temperature increases the growth rate since it increases the reaction rate of all chemical reactions within the cell. High temperature can also lead to a change in the composition and morphology of existing SEI [Palacin.2018]. Secondly mechanical stress can lead to the braking and subsequent reformation. This is dependent on the charging speeds and flexibility of the SEI. And lastly the aforementioned crossing over of cathode metals that can influence the chemical reactions within the SEI.

The SEI layer also has a profound impact on the plating behavior of cells. The thickness and ion conductivity of the layer directly influences the potential drop at the anode surface, which is the main driving force of the plating process. The mechanical properties of the layer can influence the morphology and shape of plating, i.e. whether the plated lithium metal forms as a homogeneous layer or in the form of located dendrites, that can cause potential risk to the cell safety [Bieker.2015].

Lithium Diffusion

The lithium diffusion inside of the graphite is one of the key drivers on lithium plating. Within the cell there are two different diffusion processes for the lithium ions. One is the diffusion through the electrolyte, which is in a magnitude of 0.4 to $1,4 \cdot 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ [Lee.2002]. The second part of the diffusion is the solid state diffusion in the graphite particles occurring when the concentration of lithium ions is non-homogeneous within the anode. This happens whenever ions are removed from, or added to the anode, i.e. during charging and discharging, since the transfer of the ions from the electrolyte to the anode is faster than the distributing of the ions inside of the active material. The ions are capable of moving along the vertical direction between the graphite lattice layer since the bonding between the layers is comparably weak, leaving a gap of 3,35 Å in which the ions can move and intercalate [Li.2019]. Since the magnitude of the solid state diffusion is much lower than the value for the electrolyte [Kulova.2006] the diffusion inside of the graphite is one of the limiting factors to reach high charging speeds and also the key factor for the occurrence of lithium plating. The value of the diffusion for the solid state phase

varies strongly from publication to publication, spanning everything from $10 \cdot 10^{-16}$ to $10 \cdot 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ [Kulova.2006, Persson.2010]. The main cause of this big difference stems from the fact that the diffusion rate is dependent on multiple different parameters.

Firstly, the diffusion is dependent on the temperature, with lower diffusion rate at lower temperatures, that can be described by the Arrhenius equation [Kulova.2006]. The diffusion of lithium ions starts at a temperature of 230 K [Umegaki.2017]. Secondly, it is also dependent on the SOC of the anode, i.e. the amount of lithium that is already intercalated into the [Malifarge.2018], although this effect is described to be of lower influence than the effect of the temperature [Yu.1999, Levi.1996]. This makes sense as with a higher intercalation depth, the mobility of the lithium-ions is hindered by lithium-lithium interactions [Persson.2010]. Furthermore, the distribution of the lithium-ions inside the graphite is not smooth but rather in fixed intercalation levels, called stages. These stages change abruptly when the lithium concentration reaches a certain threshold and are visible in the charge and discharge curves in form of flat plateaus. The intercalation levels are the following [Nowak.2014]:



The evolution of these stage structures also influences the diffusion coefficient [Li.2019]. Another point is that the diffusion along the edge planes is much higher than along the basal planes [Persson.2010]. Thus the edge to basal rate, particle size and particle orientation of the graphite have a big impact on the diffusion rate as well. This is also true for defects and along the grain boundaries. Here the diffusion is also severely hindered [Persson.2010]. Lastly, these parameters are also interconnected, with Umegaki et. al [Umegaki.2017] showing that the temperature dependency is different for different intercalation stages.

All these influencing factors show that the diffusion coefficient can be vastly different even within the same cell, when particle orientation, size, local temperature, local defects, can differ. Thus the lithium plating process, directly affected by the diffusion coefficient, is highly localized on the anode surface and requires a spatially resolved analysis for deeper understanding.

2.1.3 Cathode

The most commonly used metals used in the cathode are nickel, manganese and cobalt. Since all of these metals have different beneficial effects on the cell performance, a mixture of them can create a good compromise that can take the best aspects of all the possible metals. Thus most cells today use a $\text{Li}(\text{Co}_{\frac{1}{3}}\text{Ni}_{\frac{1}{3}}\text{Mn}_{\frac{1}{3}})\text{O}_2$ mixture, where the fractions of the used materials can vary. These cells are called nickel-manganese-cobalt (NMC)-cells. Research is aiming at reducing the amount of cobalt in the cells, since it is not only the most expensive of the used materials

in the cell, it is also mined under questionable conditions for the workers as well as the environment [Zubi.2018]. Apart from these metals there are also lithium-iron phosphate cells that are cheaper, have a long lifetime and comparably low environmental impact [Deng.2015], but the downside of a lower energy density, making them less suitable for mobile applications.

Because of the fact, that the anode is oversized compared to the cathode, the cathode is never fully charged again after the initial cycle [Palacin.2018].

Cathode Ageing

While the cathode of a LIB also plays a major role for the performance of the cell, the effect of the cathode on cell ageing and especially lithium plating is low compared to the anode. Lithium plating does happen on the cathode side since the potential of the cathode is very far away from the potential of metallic lithium. One noteworthy ageing mechanism, that also affects the SEI growth on the anode side of cell and thus can have an impact on lithium plating is the transition metal dissolution and migration [Palacin.2018, Lin.2015, Agubra.2013, Vetter.2005]. Here the metal ions of the cathode can be dissolved into the electrolyte and then move over to the anode surface, where they can take part in the chemical processes at the SEI and change the structure, morphology and growth rate of the SEI, often leading to a higher capacity loss and self discharge. This effect is most pronounced with cathodes containing manganese and has a considerably lower impact with cobalt-nickel mixoxides [Amatucci.1996, Liu.2003]. On the cathode surface an SEI similar to the anode SEI can form at end of charge [Palacin.2018]. The impact on the cell performance of the cathode layer however is much lower on the cell performance. This layers however can be isolating and thus lead an in-homogeneous current distribution in the cathode [Agubra.2013].

The cathode is also similarly to the anode subject to mechanical stress during the insertion of lithium ions, which can cause crystal distortion [Lin.2015].

$$A = b \cdot V_t$$

2.1.4 The Lithium Plating Process

2.1.5 The Lithium Plating Detection

XPS

The high energy of an XPS beam spot could alter the chemical composition of the SEI [Bhattacharya.2012].

3.1 XPS Measurements

Three battery samples have been prepared for the X-ray photoelectron spectroscopy (XPS) measurements for post mortem analysis. All the samples came from Sony US18650VTC3 round cells. The cells have a capacity of 1,6 A h and are operated in a voltage range from 2,5 V to 4,2 V. With a maximum discharge current of 30 A these cells can be categorized as high power cells, which is beneficial for any plating experiments at low temperatures since the cells have a comparably low internal resistance allowing for higher charge currents. Three cells have been prepared under different conditions. Two of the cells where only cycled once to bring them into the desired state, these are technically new cells. One of the cells was opened at 100% SOC (furthermore referenced as sample A), the other cell was opened at 0% SOC (sample B). The third cell was cycled for ten full cycles at -20°C to produce lithium plating in the cell and then opened (sample C). All samples from the opened batteries have been taken from the outermost area of the jellyroll. The last analyzed sample was taken from pristine anode material that was not used inside a cell (sample D). This sample was used to get a reference of the graphite surface without any electrolyte or SEI.

Sample	Description	Cycles
A	Fully charged anode	1
B	Fully discharged anode	1
C	Plated anode	10
D	Prestine anode (no SEI)	-

Table 3.1: Anode samples for XPS measurements

It has to be kept in mind that the pristine graphite is not from the same manufacturer as the anode samples from the opened batteries and is thus only used as a reference for the C1 peak and not further used in the interpretation of the results. All cells where opened inside a MBRAUN Unilab glove box under argon protective atmosphere with < 10 ppm of moisture and oxygen concentration and then transferred to the XPS device inside a gas tight transport box. The XPS device has an attached mini glove box that was used to then transfer the samples from the transport box to the XPS test chamber. Within the mini glove box the samples have

been exposed to a roughly 100 ppm oxygen atmosphere for a few minutes before entering the high vacuum. It was unfortunately not possible to also measure the moisture content in the mini glove box during the transfer.

For the XPS measurements a XPSDEVICE MISSING device was used with a base pressure of XXX Torr. The photo-electron source is a monochromatic Al K α X-ray source ($h\nu = 1486,7$ eV).

3.1.1 Preliminary Test: Influence of Sample Washing

Washing the samples taken from the battery with an appropriate solvent is often done prior to sensitive experiments. This is done to remove any battery electrolyte residue or electrolyte salt rests from the surface of the samples [Yang.2010]. The solvents used for washing are often the same or similar to the solvents used in the electrolyte (e.g. DMC). Mussa et al. however specifically mentions that washing can dissolve parts of the SEI and does not recommend washing the samples after removing them from the battery. In this case washing can falsify the measurement results since the layer is not in the original sample state anymore. Somerville et al. [Somerville.2016] did a study on the effects of washing on the surface film of graphite electrodes. They report that some authors could detect a higher sensitivity to atmosphere after the washing, which could be explained by the removal of a passivating surface film. The paper itself focused on the ratio of VC in the electrolyte, an additive commonly used in LIBs to improve the formation of a stable SEI. They postulate that the washing removes the salt and the low boiling point solvents present in the surface layer (in this case EC). Samples with different VC ratios from 0 to 6 vol.% have been prepared by them and analyzed using XPS, scanning electron microscopy (SEM), attenuated total reflectance (ATR) and fourier transform infra-red spectroscopy (FTIR). They found two different films dependent in the VC concentration with the two films showing a different SEM image as well as different behaviour in the material composition. The low VC concentration film in contrary to the higher concentration film is also very sensitive to the washing process and for this film not only the salt deposits are removed but also LiF and Li_xPF_y which can only be found in the deeper layers of the SEI. Their results show clearly that whether washing it useful or not strongly depends on the composition of the electrolyte.

For the experiment in this work commercial cells have been used, so it is not possible to obtain any data about the composition of the electrolyte since battery producers generally do not make their internal electrolyte recipes public to the competition. For this reason it could not be determined beforehand whether washing the electrodes before testing is beneficial or harmful. So in the first test run measurements of one washed and one non-washed sample from each battery anode were analyzed and the results have been compared. The samples have been washed in two DMC baths for about 30 seconds each and then dried for one hour in the glovebox before they were mounted to the sample holder and transferred to the XPS device.

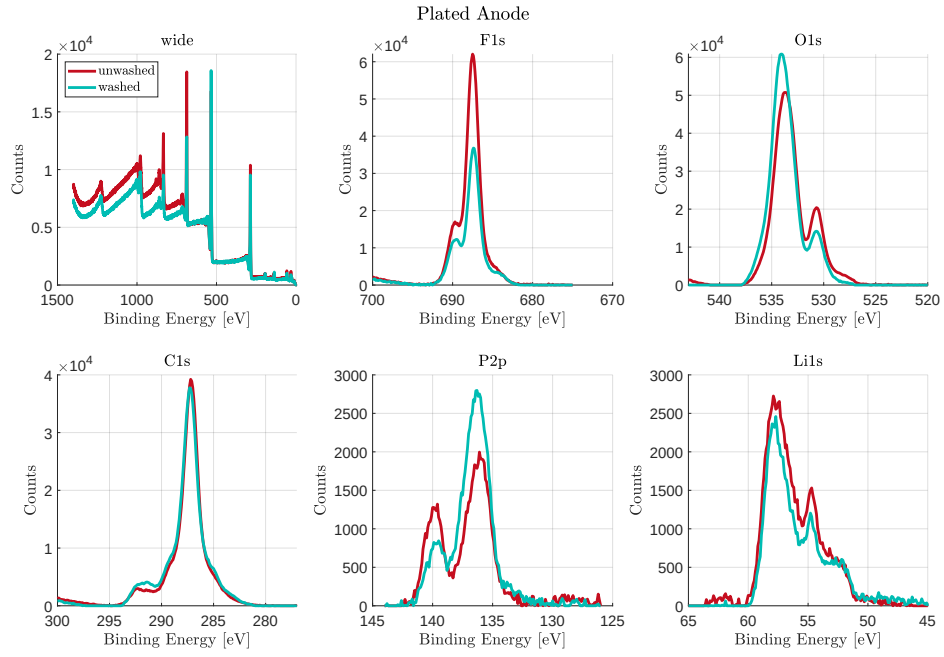


Figure 3.1: XPS spectra of the anode after 10 cycles of plating (Sample C). Comparison between the washed and unwashed sample

Chapter
Analysis

4

Chapter
Conclusion

5

Chapter **Outlook**

6